

TOPICAL REVIEW

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Optimizing Acute Ischemic Stroke Systems of Care: Current State and Future Directions

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ABSTRACT: Despite significant progress in hospital quality initiatives and the organization of regional stroke systems of care, a significant gap persists between the number of patients eligible for acute ischemic stroke reperfusion therapies and those who receive them in a timely manner. This gap reflects persistent delays and variability across the prehospital and interhospital phases of care, from prehospital dispatch and field assessment to destination selection and interhospital transfer workflows. This expert narrative review synthesizes current evidence on acute stroke systems of care, with a particular focus on prehospital identification of stroke, destination decision-making for suspected large-vessel occlusion, and interhospital transfer for patients requiring endovascular thrombectomy. Key processes, including prehospital response and scene times, destination decision-making, and door-in-door-out metrics, are examined to illustrate how system-level bottlenecks affect access to and outcomes of reperfusion therapies. Emerging and novel technologies are also described, including mobile stroke units, blood-based biomarkers for prehospital stroke diagnosis, portable neuroimaging modalities, physiological and device-based detection systems, and cross-cutting tools such as telestroke, all aimed at shifting accurate diagnosis and treatment decisions earlier in the stroke care pathway.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: biomarkers ■ critical pathways ■ hospitals ■ ischemic stroke ■ reperfusion

Stroke is a leading cause of both mortality and disability worldwide.¹ Reperfusion therapies for acute ischemic stroke (AIS) are among the most efficacious therapies in modern medicine. Intravenous thrombolysis has a number needed to treat ranging from 3.6 to 5.9 in the standard treatment window,² and endovascular thrombectomy (EVT) has a number needed to treat of 2.8 to 7.4.³ Yet, these acute reperfusion therapies are highly time-dependent, with the odds of a good functional outcome decreasing markedly with elapsed time since stroke symptom onset.^{4,5} Acute stroke systems of care have proliferated in the past 2 decades to support the timely administration of these therapies. Large nationwide quality improvement efforts have been implemented within stroke systems, for example, Target Stroke by the American Heart Association (AHA), which led to a substantial increase in rates of thrombolysis, timeliness

of treatment, and improved outcomes.⁶ These successes represent major advances for the field and have utilized systems-based approaches by implementing best practices and reducing variation in care.

Yet, key gaps remain, which are active areas of ongoing research to further optimize acute stroke systems of care. First, the proportion of eligible patients receiving timely reperfusion therapies remains suboptimal.⁷ This is a complex problem with contributions from community-level barriers to stroke symptom recognition and timely utilization of existing prehospital resources.^{8,9} Although not the primary focus of this review, community-level efforts to improve stroke symptom recognition and prompt action are vitally important. Second, prehospital stroke care processes are highly variable,¹⁰ and the optimal prehospital routing strategy in diverse care settings (eg, dispatch to

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the closest hospital versus bypass to a thrombectomy-capable center) remains unknown. Novel approaches pairing clinical prehospital assessment with blood-based biomarkers, portable neuroimaging/mobile stroke units (MSUs), and ambulance-based telestroke assessment may lead to more refined prehospital triage and thereby more precise routing strategies in the future. Even as the field seeks to optimize prehospital triage systems, interhospital transfer of stroke patients remains common,¹¹ and delays are substantial.¹² In this expert narrative review, we summarize recent updates in the literature on acute stroke systems of care, with a particular emphasis on future promising directions to optimize timely treatment.

PREHOSPITAL IDENTIFICATION OF STROKE

Call Centers and Dispatch

Given that time is brain for acute stroke treatment, systems have developed a series of critical process steps to support the timely initiation of reperfusion therapies (Figure). The first formal point of entry into the medical system is typically via ambulance or paramedic services, and the first step along this “Chain of Survival” is the phone call between the dispatch provider and the stroke patient, family member, or bystander. AHA Guidelines recommend this conversation take less than a minute,¹³ and contemporary national data suggest this time benchmark is being met in the majority of cases (median [interquartile range], 1 minute [0.5–2]).¹⁴ This is a pivotal conversation, from which all other subsequent steps cascade. The dispatcher is responsible for determining the appropriate transport type (eg, advanced life support versus basic life support ambulance, air transport, or MSU in areas where available). Although correct identification of stroke at this stage is associated with faster downstream prehospital care intervals¹⁵ and reperfusion metrics,¹⁶ incorrect identification may conversely lead to substantial delays. Most prior studies have found that stroke is correctly identified approximately half of the time at this initial dispatch step,^{15,17} even when conventional stroke screening tools are used. Authors have recognized the potential for artificial intelligence tools to improve the accuracy of stroke diagnosis in the prehospital setting,¹⁸ beginning with this first point of prehospital dispatch.^{19,20}

Prehospital Field Assessment

After dispatch, the prehospital unit responds to and arrives at the scene for their initial assessment. As with the dispatch time interval, AHA Guidelines recommend benchmarks for ambulance response times to scene arrival (<8 minutes) and scene time (<15 minutes).¹³ A

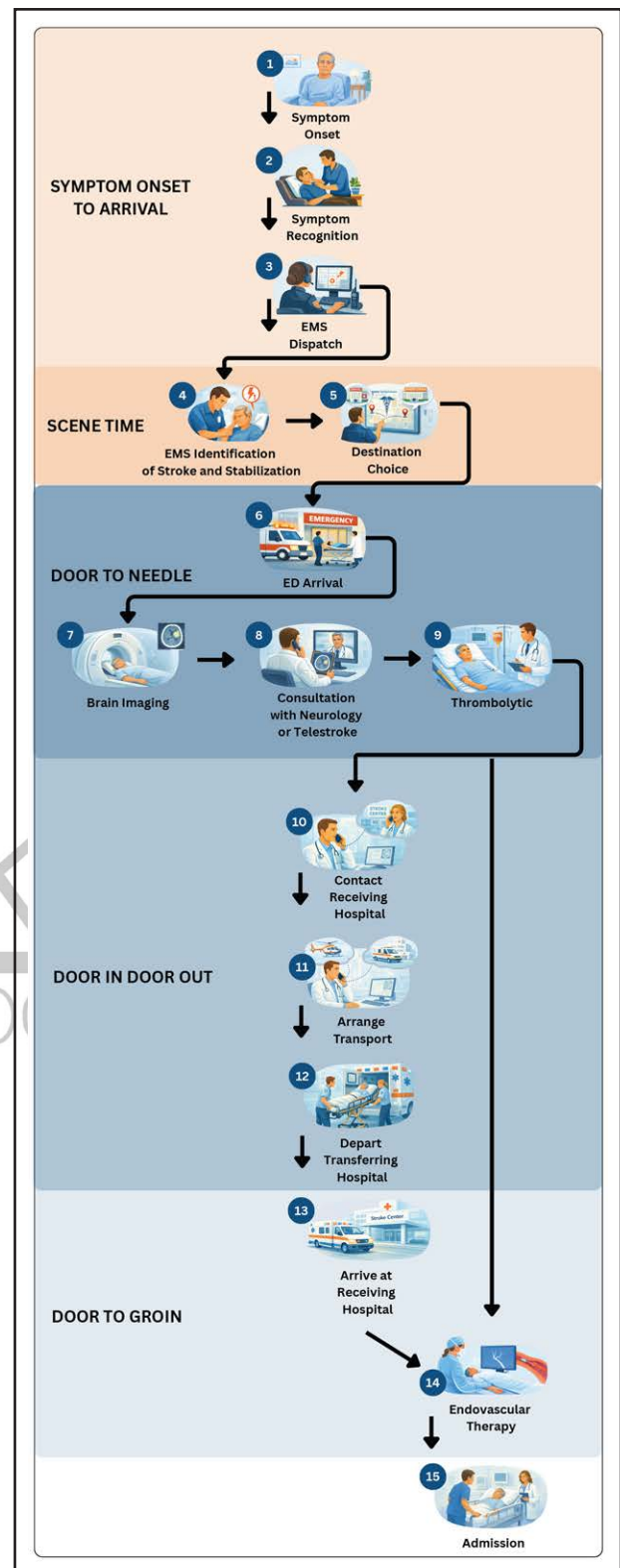


Figure. Acute stroke care cascade and key decision points.

Portions of the illustrations in this figure were generated by ChatGPT 5.5.

study using data from the National Emergency Medical Services Information System found that in the majority of cases, these guideline recommendations are being met

for ambulance response time to scene arrival (median [interquartile range], 7 minutes [5–10]).¹⁴ For ground transports, scene time has been found to be the largest component of the overall prehospital time.¹⁴ To maximize the efficiency and accuracy of on-scene assessments, prehospital stroke scales have been developed and are recommended.^{21,22} A systematic review from 2019 reviewed 23 studies on 8 commonly used prehospital stroke scales and found that the Cincinnati Prehospital Stroke Scale had the highest sensitivity for field assessments.²³ In the era of EVT, prehospital stroke screening has become more refined with the goal of selecting patients with potential large-vessel occlusion (LVO) in need of emergent routing to EVT-capable facilities. A variety of prehospital stroke severity/LVO screening tools have been developed (eg, Los Angeles Motor Scale, 3-Item Stroke Scale, Rapid Arterial Occlusion Evaluation Scale, among others). This topic has been extensively reviewed previously,²⁴ and several prior studies have directly compared these prehospital severity/LVO scales.^{25–27} While these prehospital LVO scales have fairly high sensitivity and specificity (ranging from $\approx 60\%$ – 80%),^{24,26,27} none is perfect. In a prospective observational study that compared 8 prehospital LVO scales, the area under the receiver operating characteristic curve was highest for the Rapid Arterial Occlusion Evaluation Scale but was overall similar to that of the other scales.²⁶

Emerging and Novel Technologies for Field Diagnosis and Early Treatment

Given the current imprecision and variability in the performance of prehospital stroke screening scales, various innovative strategies have emerged as potential solutions for earlier and more accurate diagnosis. These include MSUs and portable imaging technologies, surface/physiological detection systems, blood-based biomarkers, and other technologies. Differentiating hemorrhagic from ischemic stroke is the major hurdle to the hyperacute initiation of stroke therapies in the field. Historically, this determination could only be made by neuroimaging (typically computed tomography [CT]) at the hospital. These technological innovations all seek to achieve more timely and accurate stroke diagnosis and differentiation of stroke type in the prehospital setting.

MSUs are specialized ambulances equipped with the diagnostic and therapeutic capabilities to achieve hyperacute stroke treatment in the prehospital setting. These ambulances have on-board CT scanning, reperfusion therapies (eg, intravenous thrombolytics), and specialized providers (either in person or via telemedicine) to facilitate rapid reperfusion before hospital arrival. The MSU was first conceptualized in Germany in 2003²⁸ and successfully realized there in 2008.²⁹ Since that time, evidence has emerged from 2 prospective

nonrandomized controlled trials demonstrating increased rates of thrombolytic therapy, shorter last known well-to-treatment times, improved outcomes at 90 days, and similar safety profiles.^{30,31} Real-world evidence from the largest US stroke registry (Get With The Guidelines [GWTG]-Stroke) similarly found that MSU management was associated with improved functional outcomes compared with standard care.³² While multiple studies from a variety of countries and contexts have found MSUs to be cost-effective,^{33–35} funding mechanisms vary by country. Whereas some countries fund MSUs through government-based/public health mechanisms, the vast majority of MSUs within the United States and many other countries are primarily funded through philanthropy and grants.³⁶ Authors have recognized that, for MSUs to move into the mainstream globally, insurance payers will ultimately need to cover the costs.³⁷ The PRE-hospital Stroke Treatment Organization is an international consortium working to support advocacy, further research, and policy for MSU implementation.³⁸

Although MSUs show promise as a model to expedite prehospital stroke diagnosis and treatment, their use has thus far been limited by cost, lack of insurance coverage, and contextual considerations. An alternative emerging technology to differentiate hemorrhagic and ischemic stroke in the prehospital setting is blood-based biomarkers. For example, GFAP (glial fibrillary acidic protein) is an astrocyte intermediate filament protein that is rapidly released by damage to glial cells in intracerebral hemorrhage (ICH), whereas blood levels are lower in healthy controls and patients with AIS. In a recent study, investigators from Germany evaluated GFAP levels from prehospital blood samples and derived a cutoff of 55 pg/mL for plasma GFAP in the identification of ICH, with an area under the curve of 0.88 for the differentiation of ICH from AIS. They also derived a cutoff value of 30 pg/mL to exclude ICH with a negative predictive value of 97.8%.³⁹ A key gap currently remains in that this prior study used plasma blood to measure GFAP, which was drawn in the prehospital settings but centrifuged and analyzed at the hospital, limiting its applicability to hyperacute care. Within the past several months, a point-of-care assay measuring GFAP and ubiquitin carboxy-terminal hydrolase-L1 has been cleared by the US Food and Drug Administration to rule out traumatic ICH in patients presenting with traumatic brain injury.^{40–42} This test runs on a handheld device and provides results within 15 minutes. Future studies may evaluate the utility of using these point-of-care assays to differentiate AIS and ICH in the prehospital setting.

Other emerging portable diagnostic methodologies show early promise for eventual integration into prehospital systems of care. These include use of electroencephalography,^{43,44} transcranial Doppler,^{45,46} accelerometers,⁴⁷ near-infrared spectroscopy,⁴⁸ low-field portable magnetic resonance imaging,^{49,50} N20 somatosensory evoked

potentials,⁵¹ among others. This general topic has been covered previously in detail,^{24,48} so we will focus here briefly on new developments within the field. A prospective, multicenter, prehospital study published in 2023 evaluated the diagnostic accuracy of dry-electrode electroencephalography for prehospital LVO detection and found an area under the curve of the θ/α ratio of 0.80.⁴⁴ Although this diagnostic accuracy was promising, the electroencephalography data were too poor in quality for approximately a third of the study patients,⁴⁴ representing a major hurdle to widespread implementation at present. Along similar lines, a recent study found that in-hospital N20 somatosensory evoked potential responses were predictive of functional outcomes after EVT. Future applications may include their study in the prehospital setting for triage of patients with suspected LVO.⁵¹ Advances in low-field, portable magnetic resonance imaging technology have recently demonstrated the feasibility of neuroimaging in complex patient care settings,⁴⁹ with the ability to successfully detect ICH and differentiate it from AIS.^{50,52} Although CT imaging is typically used for this purpose in the acute setting, magnetic resonance imaging does offer some distinct advantages, including the lack of radiation and substantially higher sensitivity for acute ischemia. As the technology continues to improve, this may be yet another future application for prehospital stroke diagnostics.

Destination Decision-Making

In the era of EVT, an important systems question has emerged: What is the optimal prehospital routing strategy for a suspected stroke patient with potential LVO? The so-called mothership approach involves direct transport of the suspected LVO patient to a comprehensive stroke center or thrombectomy-capable hospital, bypassing a potentially nearer and less-resourced hospital (primary stroke center, acute stroke-ready hospital, or noncertified facility). Conversely, the drip-and-ship model refers to a system whereby the patient is transported to the nearest hospital for initial treatment and stabilization and then transferred for consideration of EVT. The conceptual rationale for the mothership approach is straightforward: reduce time to thrombectomy for suspected patients with LVO. Indeed, multiple systematic reviews and meta-analyses (of predominantly observational data) found an association between direct transport to a thrombectomy-capable center and improved functional outcomes.^{53–55} Yet, there is currently only 1 randomized controlled trial that has compared these 2 models of care head-to-head, and it found no significant difference in functional stroke outcomes at 90 days.⁵⁶ This study—the RACECAT trial—randomized suspected stroke patients with a Rapid Arterial Occlusion Evaluation Scale score of ≥ 5 in Catalonia, Spain. Though this trial is one of the most pivotal in the field of stroke systems, the generalizability

of the findings to other settings may be problematic. Key workflow metrics were highly optimized in this trial (median door-to-needle time of 33 minutes, median door-in-door-out [DIDO] time of 78 minutes), whereas a recent study from the US GWTG-Stroke national registry found a median DIDO time of 174 minutes among all stroke patients.¹² Furthermore, secondary and post hoc analyses of the RACECAT trial suggest that the primary results may differ based on stroke subtype (eg, ICH versus AIS)⁵⁷ and regional hospital characteristics.⁵⁸ Conducting a trial similar to RACECAT in other settings may be logistically challenging given highly variable prehospital systems and policies,¹⁰ but it would be supremely valuable, as there is compelling evidence to suggest that the optimal routing strategy may vary based on a variety of contextual factors.

INTERHOSPITAL TRANSFER AS AN ENDURING COMPONENT OF STROKE CARE

A 2019 study using data from the AHA GWTG-Stroke registry found that $\sim 43\%$ of patients receiving EVT required interhospital transfer (IHT). Furthermore, the authors found that the proportion of IHT increased over the study period (January 2012 to December 2017).¹¹ Similarly, data from the OPTIMISE registry (Optimizing Patient Treatment in Major Ischemic Stroke With EVT) in Canada found that half of all EVTs performed were in patients transferred from a primary stroke center.⁵⁹ Although improved prehospital triage and diagnostics aim to reduce the overall proportion of IHTs, the data suggest that IHT will likely remain an enduring component of acute stroke care. Indeed, with the ever-expanding indications for EVT (eg, large core)⁶⁰ and extended time windows for treatment,^{61,62} it is likely that IHTs will continue to rise steadily.

Time Metrics and Outcomes in Transferred Patients

Although IHT for EVT is increasingly common, contemporary data suggest that IHT is associated with significant delays in EVT initiation. For example, a 2019 study using GWTG-Stroke data found that transferred patients had significantly longer last-known-well to EVT initiation times compared with directly admitted patients (289 minutes versus 213 minutes; absolute standardized difference, 67.3).¹¹ Findings were similar from the Canadian OPTIMISE registry (322 minutes versus 181 minutes).⁵⁹ to localize the sources of care delays for IHT patients, several epochs of care can be considered: symptom onset-to-door time, DIDO time, interfacility transport time, and door-to-groin (DTG) time at the receiving hospital (Figure). Symptom onset-to-door time is covered

in the preceding sections (Prehospital Identification of Stroke and Destination Decision-Making).

DIDO time, defined as the time from initial ED arrival to discharge, has emerged as a major stroke quality metric and area of opportunity for patients with IHT. The 2026 AHA/American Stroke Association AIS Guidelines recommend that hospitals and prehospital professionals establish systems to prioritize reducing DIDO times for stroke transfers.⁶³ Various recommended DIDO time targets have emerged. The Joint Commission⁶⁴ and Brain Attack Coalition⁶⁵ recommend a DIDO time of <120 minutes for all stroke patients. The AHA/American Stroke Association recommends a DIDO time of ≤90 minutes for potential EVT candidates.⁶⁶ Australian guidelines recommend an even more ambitious goal of ≤60 minutes.⁶⁷ Although contemporary DIDO times vary widely,^{68–70} the largest national study using data from GWTG-Stroke found that current US DIDO times far exceed the recommended targets. In this study, the overall median DIDO time for all stroke patients was 174 minutes (interquartile range, 116–276 minutes) and was 132 minutes (interquartile range, 97–189 minutes) for EVT transfers.⁷¹ These findings suggest that DIDO time is a major area for improvement in optimizing IHT care efficiencies.

Given the clear association between shorter DTG times and better clinical outcomes,⁷² the AHA/American Stroke Association Target: Stroke Phase III initiative recommends that at least 50% of eligible patients receive DTG times of ≤90 minutes (among direct arrivals) and ≤60 minutes (among transfers).⁷³ A recent study analyzed data from GWTG-Stroke from 2017 to 2022 to identify Target: Stroke Phase III strategies associated with faster DTG times. The authors found that among directly arriving patients, alerting the neurointerventional team based on prehospital prenotification and performing CT/CT angiography (CTA) in all patients presenting within 24 hours of last known well were both independently associated with faster DTG times. Among transferred EVT patients, increased use of stroke screening tools and the use of a camera during telestroke consultation were both independently associated with shorter DTG times.⁷⁴

Studies have consistently demonstrated that transferred stroke patients have worse functional outcomes compared with directly presenting patients.^{11,75} Although some of these differences may be explained by inherent patient characteristics (eg, higher stroke severity in the transfer group),^{75,76} modifiable factors, namely transfer delays,¹¹ likely play a major role. Several single-center/system studies have found an association between prolonged DIDO time and worse functional outcomes.^{68,77} More recently, a GWTG-Stroke registry study of >20 000 patients transferred for EVT found that DIDO time delays were associated with lower odds of receiving EVT, greater odds of complications after reperfusion therapy, worse functional status, and lower

odds of independent ambulation at discharge.⁷⁸ These findings collectively support ongoing and future quality improvement initiatives to reduce DIDO times and improve outcomes for transferred patients.

Process Mapping and Failure Points

A structured risk assessment from a learning collaborative in Chicago (including 2 primary stroke centers and 3 comprehensive stroke centers) identified several key failures in the DIDO process.⁷⁹ This study used a failure modes, effects, and criticality analysis to systematically map out the DIDO process and rank the steps with the most important failures. The highest ranked process failure was failure to perform an LVO screen to detect severe stroke/LVO at the start of the DIDO diagnostic phase of care. The authors found that failure to obtain a bundled CT/CTA imaging resulted in a 45-minute increase in DIDO time. Another highly ranked failure from this study was the failure of the paramedic and ED triage nurse to perform an acute stroke screening scale, which had many downstream effects on stroke care processes. Delay in telestroke consultation initiation was also a high-ranking failure, given that in many cases this step must occur to begin the transfer process. Other more modest-risk failures included the processes of arranging transfer and communication failures between the primary stroke center's ED team and the comprehensive stroke center and its transfer center.⁷⁹

A 2024 GWTG-Stroke registry article examined DIDO process steps and found that among 28 887 patients with AIS transferred for AIS treatment or monitoring, the initial brain imaging-to-door-out interval made up 89% of the overall DIDO time, whereas the door-to-imaging interval accounted for only 11%.⁸⁰ National stroke guidelines and quality improvement initiatives have traditionally focused on improving stroke process measures early in the ED care continuum, for example, door-to-imaging time.^{13,81} Although these early care intervals are important—as everything else cascades from them—this GWTG-Stroke analysis suggests that there is presently an urgent national need to expedite care processes after initial brain imaging has been completed for transferred stroke patients.⁸⁰ Research is ongoing to optimize various processes involved, including bed availability and transport coordination with ambulance/paramedic and flight companies.

Readiness of the Receiving Hospital

Once the EVT-eligible transfer patient reaches the receiving hospital, there are additional sources of delay to reperfusion. Key elements of receiving hospital readiness include having stroke response teams that can be activated before patient arrival, standardized stroke order sets, and protocols for door straight to CT imaging to

expedite evaluation.⁸² Retrospective studies suggest that nearly a third of transferred EVT-eligible patients may receive repeat brain imaging upon arriving at the receiving hospital.⁸³ Although repeat imaging may sometimes be clinically indicated, prior literature has identified the brain imaging-to-angio-suite (picture-to-suite) interval as a major barrier to expeditious DTG times.⁸⁴

Direct to Angiosuite

As a result, several studies have recently tested whether a direct transfer to the angiosuite (DTAS) protocol may lead to faster DTG times.^{85–87} The ANGIOCAT trial, published in 2021, was a single-center randomized trial that assigned patients with LVO within 6 hours after symptom onset to either a DTAS protocol or a conventional workflow. Compared with the conventional workflow, the DTAS protocol was found to increase the odds of EVT, decrease workflow times, and improve functional outcomes as measured by the 90-day modified Rankin Scale score.⁸⁶ Of note, this study included both directly admitted and transferred patients (in a 1:2 proportion).⁸⁶ A systematic review and meta-analysis published in 2022 similarly found that DTAS was associated with faster time metrics and improved functional outcomes, with similar safety compared with the conventional approach.⁸⁷ These findings contrast with those of the recently published DIRECT ANGIO trial in France, which found that patients with acute severe neurological deficits highly suggestive of ischemic stroke due to LVO who were randomized to DTAS were at increased risk for symptomatic ICH. This study was stopped early for safety reasons without evidence of benefit in 90-day outcomes.⁸⁸ However, additional multicenter randomized controlled trials are ongoing,⁸⁹ which will provide further evaluation of the DTAS approach.

Quality Improvement Initiatives: DIDO and HI-SPEED Trial

Prior quality improvement research on DIDO time offers some potential solutions to tackle transfer delays in acute stroke care. In a simulation model of a stroke system of care, we showed that DIDO time has the second-largest impact on AIS treatment rates after the accuracy of a prehospital stroke screening scale.⁹⁰ Others have noted that DIDO time represents the single largest modifiable contributor to onset-to-reperfusion time for patients with AIS undergoing EVT.⁶⁸ In the E-SPEED study (Enhancing Stroke Prehospital Emergency Evaluation and Delivery), researchers applied a user-centered design approach to create several risk-informed solutions, which included electronic health record templates to document AIS and LVO stroke scales; electronic health record decision support to bundle CT/CTA orders based on screening scale scores; direct-to-CT and CTA protocols; standardized

location and testing of the telestroke system and ensuring that back-up telestroke neurologists were in place; and a standardized ambulance hand-off tool. A simulation study showed that combining these solutions into a single intervention could achieve a 37% (67-minute) reduction in DIDO time.⁹¹ Other solutions include prehospital prenotification to referring hospitals,¹² push-and-go thrombolysis before transfer using tenecteplase,⁹² and earlier activation of and access to ambulance services.⁹³ Another novel concept flips the paradigm by sending the interventional treating team to the patient (physician transfer instead of patient transfer) and has shown promising results.⁹⁴ The ongoing National Institutes of Health-funded cluster-randomized clinical trial, HI-SPEED (Hospital Implementation of a Stroke Protocol for Emergency Evaluation and Disposition),⁹⁵ is testing whether a multicomponent intervention delivered at 8 large health systems in the United States can reduce DIDO time and improve functional outcomes in patients with AIS transferred for EVT. The intervention includes EHR-enabled stroke screening scales, bundled CT/CTA protocols, early assessment with telestroke, interhospital communication using apps such as RAPID.AI,⁹⁶ Target: Stroke best practices for DTN reduction, early ambulance activation, and a standardized hand-off tool for paramedics.

CROSS-CUTTING ENABLERS AND IMPLEMENTATION STRATEGIES

Telestroke as an Integrative Tool

Telestroke is an acute form of telehealth that connects a remote stroke specialist to emergency department providers in lower-resourced hospitals using videoconferencing technology. Telestroke has increased access to acute stroke care by increasing rates of reperfusion therapy, especially, but not exclusively, in rural areas.⁹⁷ Although telestroke services traditionally occur in the emergency department setting, ambulance-based telestroke is an innovative strategy to move this resource into the prehospital setting. Prior studies have demonstrated the feasibility of this approach, such as the iTREAT study (Improving Treatment With Rapid Evaluation of Acute Stroke via Mobile Telemedicine), but also highlighted potential issues of connectivity and bandwidth.⁹⁸ As technology has improved, more recent studies have demonstrated more reliable wireless connections in a variety of settings.^{99–103} For example, a community-based, cluster-randomized controlled trial in New Zealand compared the diagnostic accuracy of ambulance-based telestroke assessment versus assessment by paramedics using standard scoring tools and found the telestroke method to be feasible and superior in predicting reperfusion candidates.¹⁰⁴ In a further application of telestroke to the MSU setting, investigators from the recently completed MSU-TELEMED trial sought to test whether the

use of a telemedicine neurologist on MSUs (versus a traditional onboard neurologist) led to superior resource efficiency without leading to compromised care. This trial used a prospective, randomized, blinded end-point-controlled design¹⁰⁵ and found that among patients who received thrombolytic therapy or were transported by MSU for EVT, there was no significant difference in functional outcomes at 90 days between the 2 care models.¹⁰⁶ This literature underscores the cross-cutting nature of telestroke interventions and demonstrates their potential to improve the quality and efficiency of stroke care across the acute stroke care continuum in a variety of settings.

Data Infrastructure and Learning Health Systems

A learning health system is a system that continuously learns and improves, leveraging real-world data to inform better clinical care delivery, policies, and research. Although this terminology is less often used in stroke systems,¹⁰⁷ many of the major national quality improvement initiatives, for example, Target: Stroke by the AHA/American Stroke Association, have utilized many of these key principles.^{6,74} Data linkages between sites implementing best practices and real-world quality improvement registries containing their data are a particularly powerful means of providing continuous data feedback for iterative improvement processes.

SUMMARY AND CONCLUSIONS

Despite strong trial evidence, reperfusion therapies often fail to reach patients within the critical time windows needed for maximal benefit. Although some delays originate before patients have contact with the medical system, many of the most consequential barriers occur within modifiable system and provider workflows. These include missed opportunities in prehospital identification, inefficient triage practices for severe strokes that require higher levels of care, and prolonged interfacility transfer processes that hinder access to advanced therapies.

Key strategies to address these gaps include improving prehospital recognition through enhanced screening tools and emerging diagnostic technologies; implementing destination decision-making tools that account for regional variation and resources; and optimizing interfacility transfers by standardizing processes, reducing DIDO times, and leveraging tools such as telestroke and national quality improvement initiatives. Collectively, these approaches can meaningfully shorten critical care intervals across the stroke continuum. Broad adoption of standardized protocols, integration of technological innovations, and commitment to continuous quality improvement will be essential to narrowing the gap between

evidence and practice and ensuring timely access to effective stroke therapies for all eligible patients.

ARTICLE INFORMATION

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